

Offline Presentation Round - Participant Guidelines

1. Round Overview

Congratulations on qualifying for the **Offline Presentation Round of the ML Hackathon**.

This round is designed to evaluate how effectively you have applied **Machine Learning to a Chemical Engineering problem**. The final evaluation will consider not only your model's predictive performance, but also your understanding of the underlying chemical problem, your ML methodology, scientific reasoning, and your ability to communicate and defend your solution.

The objective is **not simply to achieve the lowest RMSE**. Participants are expected to demonstrate a meaningful connection between **Chemical Engineering principles and Machine Learning**.

2. Mandatory Submission Requirements

All shortlisted teams are required to submit their solution and presentation materials through the designated Google Form.

Deadline

28 August 2026

There will be no extension of the deadline.

Late submissions may not be considered for the presentation round.

Participants must submit:

- Final presentation PDF.
- The **exact solution corresponding to their leaderboard submission**.
- Complete/source code used to generate the submitted solution.
- Any additional files specifically requested in the Google Form.

Important

The solution submitted for the offline round **must be the same solution submitted during the competition**.

The organizing team will evaluate the submitted solution **along with the corresponding code**.

Participants should therefore ensure that:

- The submitted predictions can be generated from the submitted code.
- The code corresponds to the methodology claimed in the presentation.
- The methodology presented is consistent with the submitted solution.
- No post-competition modification is made to improve the leaderboard solution.

Any significant discrepancy between the submitted solution, code, and presentation may result in **penalty or disqualification**.

3. Presentation Format

Strict Time Limit

Maximum presentation time: 5 minutes

The time limit will be strictly enforced.

Teams should therefore focus on the most important aspects of their work rather than attempting to explain every implementation detail.

Slide Limit

Maximum: 5 slides

The recommended structure is:

Slide	Suggested Content
1. Problem	Chemical Engineering problem, objective and relevance
2. Chemical Understanding	Underlying chemistry/process, important variables and assumptions
3. ML Approach	Data processing, features, models and methodology
4. Results	RMSE/leaderboard performance, comparisons and key findings
5. Failure & Application	Limitations, possible failure modes and real-world applicability

This structure is **recommended, not mandatory**, but your presentation should address the major aspects listed above.

Presentation Rules

- Maximum presentation duration: **5 minutes**
 - Maximum slides: **5**
 - Presentation should be in **PDF/PPT format**
 - At least one member of the team must be present to present.
 - The presentation must be based on the solution submitted by the team.
 - Q&A will follow the presentation.
 - Additional slides, if included, may not be considered during the judging process.
 - Teams should be prepared to explain technical details of their solution during Q&A.
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4. What Your Presentation Should Cover

A. Chemical Engineering Problem

Clearly explain:

- What problem are you solving?
- What is the target/output?
- What process or phenomenon does it represent?
- Why is the problem important?
- What is its potential industrial or practical relevance?

The judges should be able to understand the **Chemical Engineering problem before you explain the ML model**.

B. Chemical/Process Understanding

Explain the important scientific aspects behind your problem.

You may discuss:

- Relevant chemical engineering principles.
- Physical/chemical mechanisms.
- Important input variables.
- Expected relationships between variables.
- Assumptions made.
- Process constraints or operating conditions.
- Existing/conventional approaches, where relevant.

You should be able to explain **why the variables in your dataset matter from a Chemical Engineering perspective**.

5. ML Methodology

Explain how you converted the Chemical Engineering problem into an ML problem.

You may include:

- Data preprocessing.
- Exploratory analysis.
- Feature engineering.
- Feature selection.
- Models considered.
- Final model selection.
- Hyperparameter tuning.
- Validation methodology.
- Baseline comparison.

Most importantly, be prepared to explain:

Why did you choose this particular ML approach for this particular Chemical Engineering problem?

Simply stating that a particular model gave the lowest RMSE will not be sufficient. The reasoning behind the approach is important.

6. Results and Model Performance

Present the performance of your final solution clearly.

You may include:

- Final RMSE.
- Comparison with baseline models.
- Validation results.
- Important features.
- Error analysis.
- Relevant plots or observations.

You should also be prepared to explain whether your model's predictions are **consistent with expected Chemical Engineering behaviour**.

7. Failure Analysis - An Important Part of the Presentation

Your solution should not be presented as a system that works perfectly under all circumstances.

You should think about:

"If this solution were deployed in a real Chemical Engineering system and suddenly started producing incorrect predictions, what could be the reason?"

Consider **both Chemical Engineering and Machine Learning perspectives**.

Possible Chemical/Process Issues

For example:

- Change in feed composition.
- Change in operating conditions.
- Equipment degradation.
- Catalyst degradation.
- Fouling.
- Sensor/calibration issues.
- Unmodelled physical or chemical phenomena.
- Operation outside the expected process regime.
- Non-steady-state behaviour.

Possible ML/Data Issues

For example:

- Distribution shift.
- Out-of-distribution inputs.
- Extrapolation beyond training data.
- Overfitting.
- Poor feature representation.
- Missing or noisy data.
- Data pipeline errors.
- Concept drift.
- Insufficient training data.

You are **not expected to list every possible failure mode**. Instead, demonstrate that you understand the limitations of both the **physical system and the ML system**.

A strong solution should also consider:

How would you determine whether the failure originated from the chemical process, sensors/data, or the ML model?

8. Q&A and Technical Defense

After the 5-minute presentation, the judges may ask questions about your:

- Chemical Engineering assumptions.
- Understanding of the process.
- Choice of ML model.
- Feature engineering.
- Validation methodology.
- Model performance.
- Failure cases.
- Limitations.
- Real-world applicability.

Participants should be prepared to **defend the decisions made in their submitted solution**.

The Q&A is intended to assess your understanding rather than your ability to memorize technical terminology.

9. Overall Evaluation

The final score will be calculated out of **200 marks**.

The evaluation will broadly consist of the following components:

Evaluation Component	Marks
Chemical Engineering Problem & Understanding	60
ML Solution & Methodology	45
Leaderboard / Predictive Performance	40
Scientific Validation, Interpretation & Failure Analysis	30
Presentation & Technical Q&A	25

Total

200

Important

The **Chemical Engineering component carries the highest weight.**

Therefore, the competition is not solely based on leaderboard rank. A strong submission should demonstrate a balance of:

Chemical Engineering understanding + ML expertise + predictive performance + scientific reasoning + effective communication.

10. What the Judges Will Look For

At a high level, judges will evaluate whether your team can demonstrate:

1. Understanding

Do you genuinely understand the Chemical Engineering problem you are solving?

2. Relevance

Can you explain why solving this problem matters?

3. ML Reasoning

Can you justify the ML methodology rather than simply using a high-performing model?

4. Performance

How well does your solution perform on the competition metric?

5. Scientific Validity

Do your predictions and conclusions make sense from a Chemical Engineering perspective?

6. Critical Thinking

Can you identify where your solution might fail?

7. Communication

Can you explain your approach clearly within the 5-minute limit?

8. Technical Defense

Can you justify your decisions when questioned by the judges?

11. Recommended Presentation Flow

Given the **5-minute limit**, teams may consider approximately:

0:00–1:00 → Chemical Engineering Problem

1:00–2:00 → Chemical Understanding

2:00–3:15 → ML Methodology

3:15–4:15 → Results & Interpretation

4:15–5:00 → Failure Analysis & Limitations

This is only a suggested structure. Teams may organize their presentation differently as long as the important aspects are covered.

12. Important Restrictions

Participants must adhere to the following:

1. **The submitted solution must correspond to the original competition submission.**
 2. **The submitted code must correspond to the submitted solution.**
 3. Do not introduce post-competition modifications intended to improve the submitted predictions.
 4. Do not use information that was unavailable under the competition rules to generate the submitted solution.
 5. Do not claim methodologies or experiments that were not actually performed.
 6. All results presented must correspond to the submitted work.
 7. The presentation must remain within the **5-minute limit**.
 8. The presentation must not exceed **5 slides**.
 9. Teams must be prepared to explain and defend their methodology during Q&A.
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Final Message to Participants

Your leaderboard score gets you to the final round. Your understanding of the Chemical Engineering problem, the reasoning behind your ML solution, and your ability to defend it will determine your final score.

Focus on demonstrating **why your solution works, why it is appropriate for the chemical problem, and when it might fail**-not merely on reporting your RMSE.